Trade name: 74000-00199, ink, yellow

Current version: 3.0.0, Revision: 15.12.2025 **Replaced version:** 2.0.0, Revision: 20.08.2025 **Region:** GER

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name

74000-00199, ink, yellow

UFI:
2KSG-01QT-W00S-KD45

1.2 Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses of the substance or mixture
printer's ink
Ink

Uses advised against
No data available.

1.3 Details of the supplier of the safety data sheet

Address
Paul Leibinger GmbH & Co. KG
Daimlerstasse 14
78532 Tuttlingen

Telephone no.	+49 (0)7461 9286 0
Fax no.	+49 (0)7461 9286 119
e-mail	info@leibinger-group.com

Advice on Safety Data Sheet
sdb_info@umco.de

Subsidiary USA:

Address
Paul Leibinger Inc
2702 Buell Drive, Suite B
East Troy, WI 52120 USA
Phone: +01 262 642 4030

1.4 Emergency telephone number

For medical advice (in German and English):
+49 (0)551 192 40 (Giftinformationszentrum Nord)

For medical advice in the USA (Transport & Environment):
+1 800 255 3924 (24h service)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification in accordance with Regulation (EC) No 1272/2008 (CLP)
Eye Irrit. 2; H319
Flam. Liq. 2; H225
STOT SE 3; H336

Classification information
This product is assessed and classified using the methods and criteria below referred to in Article 9 of Regulation (EC) n° 1272/2008:
Physical hazards: determined through assessment data based on the methods or standards referred to in part 2 of Annex I to CLP
Health hazards and environmental hazards: determined through toxicological and ecotoxicological assessment data based on the methods or standards referred to in Part 3, 4 and 5 of Annex I to CLP.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP Regulation)

Hazard pictograms


GHS02


GHS07

Signal word
Danger

Hazardous component(s) to be indicated on label:
butanone

Hazard statement(s)

H225	Highly flammable liquid and vapour.
H319	Causes serious eye irritation.
H336	May cause drowsiness or dizziness.

Hazard statements (EU)

EUH066	Repeated exposure may cause skin dryness or cracking.

Precautionary statement(s)

P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P240	Ground and bond container and receiving equipment.
P241	Use explosion-proof electrical/ventilating/lighting/ equipment.
P242	Use non-sparking tools.
P243	Take action to prevent static discharges.
P261	Avoid breathing dust/fume/gas/mist/vapours/spray.
P264	Wash thoroughly after handling.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P370+P378	In case of fire: Use water spray, extinguishing powder, foam or CO2 to extinguish.

UFI:
2KSG-01QT-W00S-KD45

Labelling information
Other known names for components in the present mixture:
EC 943-133-8: Mixture of Solvent Orange 54, Solvent Yellow 82 and Solvent Red 122

2.3 Other hazards

The substance/mixture contains components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

PBT assessment
The product is not considered to be a PBT.

vPvB assessment
The product is not considered to be a vPvB.

SECTION 3: Composition/information on ingredients

3.1 Substances

Not applicable. The product is not a substance.

3.2 Mixtures

Chemical characterization
Mixture based on: solvent; Resins; Cellulose nitrate; dyes

Hazardous ingredients

No	Substance name	Additional information	

	CAS / EC / Index / REACH no	Classification (EC) 1272/2008 (CLP)	Concentration	%
1	**butanone**			
	78-93-3 201-159-0 606-002-00-3 01-2119457290-43	Flam. Liq. 2; H225 Eye Irrit. 2; H319 STOT SE 3; H336 EUH066	40,00 - 60,00	wt%
2	**ethanol**			
	64-17-5 200-578-6 603-002-00-5 01-2119457610-43	Flam. Liq. 2; H225 Eye Irrit. 2; H319	>= 5,00 - < 10,00	wt%
3	**1-methoxy-2-propanol**			
	107-98-2 203-539-1 603-064-00-3 01-2119457435-35	Flam. Liq. 3; H226 STOT SE 3; H336	< 5,00	wt%
4	**ethyl-acetate**			
	141-78-6 205-500-4 607-022-00-5 01-2119475103-46	EUH066 Eye Irrit. 2; H319 Flam. Liq. 2; H225 STOT SE 3; H336	< 5,00	wt%
5	**sodium perchlorate**			
	7601-89-0 231-511-9 017-010-00-6 -	Ox. Sol. 1; H271 Acute Tox. 4; H302 Eye Irrit. 2; H319 STOT RE 2; H373o	< 5,00	wt%

Full text of H- and EUH-phrases, if not already mentioned in section 2.2: see section 16.

No	Note	Specific concentration limits	M-factor (acute)	M-factor (chronic)
2	-	Eye Irrit. 2; H319: C >= 50%	-	-

No	Route, target organ, concrete effect
5	H373o oral; -; -

3.3 Other information

Other known names for components in the present mixture:
EC 943-133-8: Mixture of Solvent Orange 54, Solvent Yellow 82 and Solvent Red 122

SECTION 4: First aid measures

4.1 Description of first aid measures

General information
Remove contaminated clothing and shoes immediately, and launder thoroughly before reusing. In case of persisting adverse effects, consult a physician.

After inhalation
Remove affected persons from dangerous area by observing suitable respiratory protection measures. Ensure supply of fresh air.

After skin contact
In case of contact with skin wash off with water.

After eye contact
Remove contact lenses. Rinse eye thoroughly under running water keeping eyelids wide open and protecting the unaffected eye (at least 10 to 15 minutes). Get medical attention if pain still persists.

After ingestion
Rinse the mouth thoroughly with water. Do not induce vomiting. Never give anything by mouth to an unconscious person.

4.2 Most important symptoms and effects, both acute and delayed

No data available.

4.3 Indication of any immediate medical attention and special treatment needed

No data available.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media
Alcohol resistant foam, CO2, powders, water spray

Unsuitable extinguishing media
High power water jet

5.2 Special hazards arising from the substance or mixture

In the event of fire, the following can be released: Carbon dioxide (CO_2); Carbon monoxide (CO); Nitrogen oxides (NO_x); chromium compounds; chlorine compounds

5.3 Advice for firefighters

Use self-contained breathing apparatus. Wear protective clothing. Containers close to fire should be transferred to a safe place. Cool closed containers exposed to fire with water.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel
Refer to protective measures listed in sections 7 and 8. Keep away from ignition sources.

For emergency responders
Personal protective equipment (PPE) - see section 8.

6.2 Environmental precautions

Do not discharge into the drains/surface waters/groundwater. Do not discharge into the subsoil/soil.

6.3 Methods and material for containment and cleaning up

Contain and collect spillage with non-combustible absorbent materials, e.g. sand, earth, vermiculite, diatomaceous earth and place in container for disposal according to local regulations (see section 13).

6.4 Reference to other sections

Information regarding safe handling, see section 7. Information regarding personal protective measures, see section 8. Information regarding waste disposal, see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling
Risks inherent to handling the product must be minimised by applying the appropriate protective and preventive measures. Working processes should - so far as possible, according to the state of the art - be designed to rule out bodily contact or the release of hazardous substances.

General protective and hygiene measures
Do not eat, drink or smoke during work time. Keep away from foodstuffs and beverages. Do not inhale vapours. Avoid contact with eyes and skin. Wash hands before breaks and after work. Remove contaminated clothing and shoes and launder thoroughly before reusing.

Advice on protection against fire and explosion
Keep away from ignition sources and provide for good ventilation. Vapours can form an explosive mixture with air. Isolate from sources of heat, sparks and open flame. Take precautionary measures against electrostatic loading (earthing necessary during loading operations). Use explosion-proof equipment/fittings and non-sparking tools.

7.2 Conditions for safe storage, including any incompatibilities

Technical measures and storage conditions
Keep container tightly closed and dry in a cool, well-ventilated place.

Recommended storage temperature

Value	10	-	25	°C

Requirements for storage rooms and vessels
Containers which are opened must be carefully closed and kept upright to prevent leakage. Always keep in containers of same material as the original.

Incompatible products
Substances to be avoided, see section 10.

Stoarge Class according TRGS 510

3	Flammable liquids

7.3 Specific end use(s)

Recommendations
Ink for industrial CIJ printers

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational exposure limit values

No	Substance name	CAS no.		EC no.	
1	**butanone**	**78-93-3**		**201-159-0**	
	TRGS 900				
	Butanon				
	WEL long-term (8-hr TWA reference period)	600	mg/m³	200	ml/m³
	Ceiling Limit Skin resorption / sensibilisation Notes	1(I) H Y			
	2000/39/EC				
	Butanone				
	WEL short-term (15 min reference period)	900	mg/m³	300	ppm
	WEL long-term (8-hr TWA reference period)	600	mg/m³	200	ppm
2	**ethanol**	**64-17-5**		**200-578-6**	
	TRGS 900				
	Ethanol				
	WEL long-term (8-hr TWA reference period)	380	mg/m³	200	ml/m³
	Ceiling Limit Notes	4 (II) Y			
3	**1-methoxy-2-propanol**	**107-98-2**		**203-539-1**	
	TRGS 900				
	1-Methoxy-2-propanol				
	WEL long-term (8-hr TWA reference period)	370	mg/m³	100	ml/m³
	Ceiling Limit Notes	2(I) Y			
	2000/39/EC				
	1-Methoxypropanol-2				
	WEL short-term (15 min reference period)	568	mg/m³	150	ppm
	WEL long-term (8-hr TWA reference period)	375	mg/m³	100	ppm
	Skin resorption / sensibilisation	Skin			
4	**ethyl-acetate**	**141-78-6**		**205-500-4**	
	2017/164/EU				
	Ethyl acetate				
	WEL short-term (15 min reference period)	1468	mg/m³	400	ppm
	WEL long-term (8-hr TWA reference period)	734	mg/m³	200	ppm
	TRGS 900				
	Ethylacetat				

	WEL long-term (8-hr TWA reference period)	730	mg/m³	200	ml/m³
	Ceiling Limit Notes	2(I) Y			

Biological limit values

No	Substance name		
1	**butanone**		
	TRGS 903		
	2-Butanon (Methylethylketon)		
	parameter	2-Butanon	
	Value	2	mg/l
	Comments sample material Sampling moment	DFG U b	
2	**1-methoxy-2-propanol**		
	TRGS 903		
	1-Methoxypropan-2-ol		
	parameter	1-Methoxypropan-2-ol	
	Value	15	mg/l
	Comments sample material Sampling moment	DFG U b	

DNEL, DMEL and PNEC values

DNEL values (worker)

No	Substance name			CAS / EC no	
	Route of exposure	**Exposure time**	**Effect**	**Value**	
1	**butanone**			**78-93-3** **201-159-0**	
	dermal	Long term (chronic)	systemic	1161	mg/kg/day
	inhalative	Long term (chronic)	systemic	600	mg/m³
	inhalative	Short term (acut)	systemic	900	mg/m³
2	**ethanol**			**64-17-5** **200-578-6**	
	dermal	Long term (chronic)	systemic	8238	mg/kg/day
	inhalative	Long term (chronic)	systemic	380	mg/m³
3	**1-methoxy-2-propanol**			**107-98-2** **203-539-1**	
	dermal	Long term (chronic)	systemic	183	mg/kg/day
	inhalative	Long term (chronic)	systemic	369	mg/m³
	inhalative	Short term (acut)	local	553,5	mg/m³
	inhalative	Short term (acut)	systemic	553,5	mg/m³
4	**ethyl-acetate**			**141-78-6** **205-500-4**	
	dermal	Long term (chronic)	systemic	63	mg/kg/day
	inhalative	Long term (chronic)	systemic	734	mg/m³
	inhalative	Short term (acut)	systemic	1468	mg/m³
	inhalative	Long term (chronic)	local	734	mg/m³
	inhalative	Short term (acut)	local	1468	mg/m³

DNEL value (consumer)

No	Substance name			CAS / EC no	
	Route of exposure	**Exposure time**	**Effect**	**Value**	
1	**butanone**			**78-93-3** **201-159-0**	
	oral	Long term (chronic)	systemic	31	mg/kg/day
	dermal	Long term (chronic)	systemic	412	mg/kg/day
	inhalative	Long term (chronic)	systemic	106	mg/m³

	inhalative	Short term (acut)	systemic	450	mg/m³
2	**ethanol**			**64-17-5** **200-578-6**	
	inhalative	Long term (chronic)	systemic	114	mg/m³
3	**1-methoxy-2-propanol**			**107-98-2** **203-539-1**	
	oral	Long term (chronic)	systemic	33	mg/kg/day
	dermal	Long term (chronic)	systemic	78	mg/kg/day
	inhalative	Long term (chronic)	systemic	43,9	mg/m³
4	**ethyl-acetate**			**141-78-6** **205-500-4**	
	oral	Long term (chronic)	systemic	4,5	mg/kg/day
	dermal	Long term (chronic)	systemic	37	mg/kg/day
	inhalative	Long term (chronic)	systemic	367	mg/m³
	inhalative	Short term (acut)	systemic	734	mg/m³
	inhalative	Long term (chronic)	local	367	mg/m³
	inhalative	Short term (acut)	local	734	mg/m³

PNEC values

No	**Substance name**		**CAS / EC no**	
	ecological compartment	**Type**	**Value**	
1	**ethanol**		**64-17-5** **200-578-6**	
	water	fresh water	0,96	mg/L
	water	marine water	0,79	mg/L
	water	fresh water sediment	3,6	mg/kg dry weight
	water	marine water sediment	2,9	mg/L
	soil	-	0,63	mg/kg dry weight
	sewage treatment plant	-	580	mg/L
	secondary poisoning	-	0,38	g/kg
	with reference to: food			
2	**1-methoxy-2-propanol**		**107-98-2** **203-539-1**	
	water	fresh water	10	mg/L
	water	marine water	1	mg/L
	water	Aqua intermittent	100	mg/L
	water	fresh water sediment	52,3	mg/kg
	with reference to: dry weight			
	water	marine water sediment	5,2	mg/kg
	with reference to: dry weight			
	soil	-	4,59	mg/kg
	with reference to: dry weight			
	sewage treatment plant	-	100	mg/L
3	**ethyl-acetate**		**141-78-6** **205-500-4**	
	water	fresh water	0,24	mg/L
	water	marine water	0,024	mg/L
	water	fresh water sediment	1,15	mg/kg dry weight
	water	marine water sediment	0,115	mg/kg dry weight
	soil	-	0,148	mg/kg dry weight
	sewage treatment plant	-	650	mg/L
	secondary poisoning	-	0,2	g/kg
	with reference to: food			

8.2 Exposure controls

Appropriate engineering controls
Provide adequate ventilation. Where reasonably practicable this should be achieved by the use of local exhaust ventilation and good general extraction. If these are not sufficient to maintain concentrations of particulates and solvent vapour below the OEL (=Occupational Exposure Limit), suitable respiratory protection must be worn.

Personal protective equipment

Respiratory protection
If workplace exposure limits are exceeded, a respiration protection approved for this particular job must be worn. In case of aerosol and mist formation, take appropriate measures for breathing protection in the event workplace threshold values are not specified. Short term: filter apparatus, Filter A2

Eye / face protection
Tightly fitting safety glasses (EN 166).

Hand protection
Sufficient protection is given wearing suitable protective gloves checked according to i.e. EN 374, in the event of risk of skin contact with the product. Before use, the protective gloves should be tested in any case for its specific work-station suitability (i.e. mechanical resistance, product compatibility and antistatic properties). Adhere to the manufacturer's instructions and information relating to the use, storage, care and replacement of protective gloves. Protective gloves shall be replaced immediately when physically damaged or worn. Design operations thus to avoid permanent use of protective gloves.

Appropriate Material	butyl rubber		
Material thickness	>=	0,7	mm
Breakthrough time	>=	60	min

Other
Normal chemical work clothing.

Environmental exposure controls
No data available.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

State of aggregation	
liquid	

Colour	
yellow	

Odour	
alcohol-like	

pH value	
reason for missing pH	substance/mixture is non-polar/aprotic

Boiling point / boiling range			
Value	>	35	°C
Source	supplier		

Melting point/freezing point	
No data available	

Decomposition temperature	
No data available	

Flash point			
Value		-4	°C
Source	supplier		

Ignition temperature	
No data available	

Flammability
The product is flammable.

Lower explosion limit
No data available

Upper explosion limit
No data available

Vapour pressure
No data available

Relative vapour density
No data available

Relative density
No data available

Density			
Value		0,973	g/cm³
Reference temperature		20	°C
Source	supplier		

Solubility
No data available

Partition coefficient n-octanol/water (log value)				
No	**Substance name**	**CAS no.**		**EC no.**
1	**butanone**	**78-93-3**		**201-159-0**
log Pow			0,3	
Reference temperature			40	°C
Method	OECD 117			
Source	ECHA			
2	**ethanol**	**64-17-5**		**200-578-6**
log Pow			-0,35	
Reference temperature			24	°C
with reference to	pH 7,4			
Method	OECD 107			
Source	ECHA			
3	**1-methoxy-2-propanol**	**107-98-2**		**203-539-1**
log Pow	<		1	
Reference temperature			20	°C
with reference to	pH: 6.8			
Method	OECD 117			
Source	ECHA			
4	**ethyl-acetate**	**141-78-6**		**205-500-4**
log Pow			0,68	
Reference temperature			25	°C
with reference to	pH 7			
Method	EPA OPPTS 830.7560			
Source	ECHA			

Kinematic viscosity			
Value	appr.	7,8	mPa*s
Type	dynamic		
Source	supplier		

Particle characteristics
No data available

9.2 Other information

Other information
VOC: 629 g/l

SECTION 10: Stability and reactivity

10.1 Reactivity
Dangerous reactions are not expected if the product is handled according to its intended use.

10.2 Chemical stability
Stable under recommended storage and handling conditions (See section 7).

10.3 Possibility of hazardous reactions
Dangerous reactions are not to be expected when handling product according to its intended use.

10.4 Conditions to avoid
Heat, naked flames and other ignition sources.

10.5 Incompatible materials
strong oxidizing agents

10.6 Hazardous decomposition products
None, if handled according to intended use.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute oral toxicity (result of the ATE calculation for the mixture)	
Product Name	
74000-00199, ink, yellow	
Comments	The result of the applied calculation method according to the European Regulation (EC) 1272/2008 (CLP), Paragraph 3.1.3.6, Part 3 of Annex I is outside the values that imply a classification / labelling of this mixture according to table 3.1.1 defining the respective categories (ATE oral > 2000 mg/kg).

Acute oral toxicity				
No	**Substance name**		**CAS no.**	**EC no.**
1	**butanone**		**78-93-3**	**201-159-0**
LD50			2054	mg/kg bodyweight
Species		rat		
Method		OECD 423		
Source		ECHA		
Evaluation/classification		Based on available data, the classification criteria are not met.		
2	**ethanol**		**64-17-5**	**200-578-6**
LD50			10470	mg/kg bodyweight
Species		rat		
with reference to		95% ethanol in water		
Method		OECD 401		
Source		ECHA		
Evaluation/classification		Based on available data, the classification criteria are not met.		
3	**1-methoxy-2-propanol**		**107-98-2**	**203-539-1**
LD50			4016	mg/kg bodyweight
Species		rat		
Method		EC 440/2008, B.1		
Source		ECHA		
4	**ethyl-acetate**		**141-78-6**	**205-500-4**
LD50		>	5600	mg/kg bodyweight
Species		rat		
Source		ECHA		

Acute dermal toxicity			
No	**Substance name**	**CAS no.**	**EC no.**
1	**1-methoxy-2-propanol**	**107-98-2**	**203-539-1**

LD50	>	2000	mg/kg bodyweight
Species	rat		
Method	440/2008/EC B.3.		
Source	ECHA		
2 ethyl-acetate	**141-78-6**		**205-500-4**
LD50	>	20000	mg/kg bodyweight
Species	rabbit		
Source	ECHA		

Acute inhalational toxicity			
No Substance name	**CAS no.**		**EC no.**
1 ethanol	**64-17-5**		**200-578-6**
LC50		124,7	mg/l
Duration of exposure		4	h
State of aggregation	Vapour		
Species	rat		
Method	OECD 403		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		

Skin corrosion/irritation			
No Substance name	**CAS no.**		**EC no.**
1 butanone	**78-93-3**		**201-159-0**
Duration of exposure		4	h
Species	rabbit		
Method	OECD 404		
Source	ECHA		
Evaluation	non-irritant		
Evaluation/classification	Based on available data, the classification criteria are not met.		
2 ethanol	**64-17-5**		**200-578-6**
Species	rabbit		
Method	OECD 404		
Source	ECHA		
Evaluation	non-irritant		
Evaluation/classification	Based on available data, the classification criteria are not met.		
3 1-methoxy-2-propanol	**107-98-2**		**203-539-1**
Species	rabbit		
Method	EC 440/2008, B.4		
Source	ECHA		
Evaluation	non-irritant		
4 ethyl-acetate	**141-78-6**		**205-500-4**
Species	rabbit		
Method	OECD 404		
Source	ECHA		
Evaluation	low-irritant		
Evaluation/classification	Based on available data, the classification criteria are not met.		

Serious eye damage/irritation		
No Substance name	**CAS no.**	**EC no.**
1 butanone	**78-93-3**	**201-159-0**
Species	rabbit	
Method	OECD 405	
Source	ECHA	
Evaluation	irritant	
Evaluation/classification	Based on available data, the classification criteria are met.	
2 ethanol	**64-17-5**	**200-578-6**
Species	rabbit	
Method	OECD 405	
Source	ECHA	
Evaluation	irritant	
Evaluation/classification	Based on available data, the classification criteria are met.	

3	1-methoxy-2-propanol	107-98-2	203-539-1
Species	rabbit		
Method	2004/73/EEC, B.5		
Source	ECHA		
Evaluation	non-irritant		
4	**ethyl-acetate**	**141-78-6**	**205-500-4**
Species	rabbit		
Method	OECD 405		
Source	ECHA		
Evaluation	low-irritant		

Respiratory or skin sensitisation			
No	**Substance name**	**CAS no.**	**EC no.**
1	**butanone**	**78-93-3**	**201-159-0**
Route of exposure	Skin		
Species	guinea pig		
Method	OECD 406		
Source	ECHA		
Evaluation	non-sensitizing		
Evaluation/classification	Based on available data, the classification criteria are not met.		
2	**ethanol**	**64-17-5**	**200-578-6**
Route of exposure	respiratory tract		
Source	ECHA		
Evaluation	non-sensitizing		
Evaluation/classification	Based on available data, the classification criteria are not met.		
Route of exposure	Skin		
Species	mouse		
Source	ECHA		
Evaluation	non-sensitizing		
Evaluation/classification	Based on available data, the classification criteria are not met.		
3	**1-methoxy-2-propanol**	**107-98-2**	**203-539-1**
Route of exposure	Skin		
Species	guinea pig		
Method	440/2008/EC B.6		
Source	ECHA		
Evaluation	non-sensitizing		
4	**ethyl-acetate**	**141-78-6**	**205-500-4**
Route of exposure	Skin		
Species	guinea pig		
Method	OECD 406		
Source	ECHA		
Evaluation	non-sensitizing		

Germ cell mutagenicity			
No	**Substance name**	**CAS no.**	**EC no.**
1	**butanone**	**78-93-3**	**201-159-0**
Type of examination	in vitro gene mutation study in bacteria		
Species	Salmonella typhimurium		
Method	OECD 471		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
Type of examination	In vitro Mammalian Chromosomal Aberration Test		
Species	rat		
Method	OECD 473		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
Type of examination	In vitro mammalian cell gene mutation test		
Species	Mouse lymphoma cells		
Method	OECD 476		
Source	ECHA		

Evaluation/classification	Based on available data, the classification criteria are not met.		
Type of examination	In vivo mammalian somatic cell study: cytogenicity / erythrocyte micronucleus		
Species	mouse		
Method	OECD 474		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
2	**ethanol**	**64-17-5**	**200-578-6**
Type of examination	in vitro gene mutation study in bacteria		
Species	Salmonella typhimurium		
Method	OECD 471		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
Type of examination	in vitro gene mutation study in mammalian cells		
Species	mouse lymphoma cells		
Method	OECD 476		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
Type of examination	Genotoxicity in vivo		
Species	mouse		
Method	OECD 478		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
3	**ethyl-acetate**	**141-78-6**	**205-500-4**
Type of examination	Bacterial Reverse Mutation Test		
Species	S. typhimurium, other: TA 1535, TA 1537, TA 97, TA98 and TA 100		
Method	OECD 471		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		

Reproduction toxicity			
No	**Substance name**	**CAS no.**	**EC no.**
1	**butanone**	**78-93-3**	**201-159-0**
Route of exposure	inhalational		
Type of examination	Prenatal Developmental Toxicity Study		
Species	rat		
Method	OECD 414		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
2	**ethanol**	**64-17-5**	**200-578-6**
Route of exposure	oral		
NOAEL			
Type of examination	2 generation study		
Species	mouse		
Method	OECD 416		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
Route of exposure	inhalational		
NOAEL	>=	20000	ppm
Type of examination	Prenatal Developmental Toxicity Study		
Species	rat		
Method	OECD 414		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
3	**ethyl-acetate**	**141-78-6**	**205-500-4**
Type of examination	Two-Generation Reproduction Toxicity Study		
Species	mouse		
Method	OECD 416		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		

Carcinogenicity			
No	Substance name	CAS no.	EC no.
1	butanone	78-93-3	201-159-0
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
2	ethanol	64-17-5	200-578-6
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		

STOT - single exposure			
No	Substance name	CAS no.	EC no.
1	ethyl-acetate	141-78-6	205-500-4
Route of exposure	inhalational		
NOEC		350	ppm
Species	rat		
Source	ECHA		
Effects	May cause drowsiness or dizziness.		
Evaluation/classification	Based on available data, the classification criteria are met.		

STOT - repeated exposure			
No	Substance name	CAS no.	EC no.
1	butanone	78-93-3	201-159-0
Route of exposure	inhalational		
Species	rat		
Method	OECD 413		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		
2	ethanol	64-17-5	200-578-6
Route of exposure	oral		
Duration of exposure		14	week/s
Species	rat		
Target organ	kidneys		
Method	OECD 408		
Source	ECHA		
Evaluation/classification	Based on available data, the classification criteria are not met.		

Aspiration hazard
No data available

Endocrine disrupting properties			
No	Substance name	CAS no.	EC no.
1	sodium perchlorate	7601-89-0	231-511-9
The substance has no endocrine disrupting properties. (human)			
Source ECHA			

11.2 Information on other hazards

Other information
No data available.

SECTION 12: Ecological information

12.1 Toxicity

Toxicity to fish (acute)			
No	Substance name	CAS no.	EC no.
1	butanone	78-93-3	201-159-0
LC50		2973	mg/l
Duration of exposure		96	h
Species	Pimephales promelas		
Method	OECD 203		
Source	ECHA		

2	**ethanol**	**64-17-5**		**200-578-6**	
LC50				14200	mg/l
Duration of exposure				96	h
Species		Pimephales promelas			
Method		EPA			
Source		ECHA			
3	**1-methoxy-2-propanol**	**107-98-2**		**203-539-1**	
LC50		> 4600	-	10000	mg/l
Duration of exposure				96	h
Species		Leuciscus idus			
Method		DIN 38 412, part L15			
Source		ECHA			
Evaluation/classification		Based on available data, the classification criteria are not met.			
4	**ethyl-acetate**	**141-78-6**		**205-500-4**	
LC50				220	mg/l
Duration of exposure				96	h
Species		Pimephales promelas			
Source		ECHA			

Toxicity to fish (chronic)
No data available

Toxicity to Daphnia (acute)					
No	**Substance name**	**CAS no.**		**EC no.**	
1	**butanone**	**78-93-3**		**201-159-0**	
EC50				308	mg/l
Duration of exposure				48	h
Species		Daphnia magna			
Method		OECD 202			
Source		ECHA			
2	**ethanol**	**64-17-5**		**200-578-6**	
EC50				5012	mg/l
Duration of exposure				48	h
Species		Ceriodaphnia dubia			
Method		ASTM Standard E 729-80			
Source		ECHA			
3	**1-methoxy-2-propanol**	**107-98-2**		**203-539-1**	
EC50		21100	-	25900	mg/l
Duration of exposure				48	h
Species		Daphnia magna			
Method		ESR-ES-15			
Source		ECHA			
Evaluation/classification		Based on available data, the classification criteria are not met.			
4	**ethyl-acetate**	**141-78-6**		**205-500-4**	
EC50				3090	mg/l
Duration of exposure				24	h
Species		Daphnia magna			
Source		ECHA			

Toxicity to Daphnia (chronic)					
No	**Substance name**	**CAS no.**		**EC no.**	
1	**ethanol**	**64-17-5**		**200-578-6**	
NOEC				9,6	mg/l
Duration of exposure				9	day(s)
Species		Daphnia magna			
Source		ECHA			
2	**ethyl-acetate**	**141-78-6**		**205-500-4**	
NOEC				2,4	mg/l
Species		Daphnia magna			
Method		OECD 211			

Toxicity to algae (acute)			
No **Substance name**	**CAS no.**		**EC no.**
1 **butanone**	**78-93-3**		**201-159-0**
EC50		1220	mg/l
Duration of exposure		96	h
Species	Raphidocelis subcapitata		
Method	OECD 201		
Source	ECHA		
2 **ethanol**	**64-17-5**		**200-578-6**
EC50		275	mg/l
Duration of exposure		72	h
Species	Chlorella vulgaris		
Method	OECD 201		
Source	ECHA		

Toxicity to algae (chronic)			
No **Substance name**	**CAS no.**		**EC no.**
1 **ethyl-acetate**	**141-78-6**		**205-500-4**
NOEC	>	100	mg/l
Species	Desmodesmus subspicatus		
Method	OECD 201		
Source	ECHA		

Bacteria toxicity
No data available

12.2 Persistence and degradability

Biodegradability			
No **Substance name**	**CAS no.**		**EC no.**
1 **butanone**	**78-93-3**		**201-159-0**
Type	aerobic biodegradation		
Value		98	%
Duration		28	day(s)
Method	OECD 301 D		
Source	ECHA		
Evaluation	readily biodegradable		
2 **ethanol**	**64-17-5**		**200-578-6**
Type	aerobic biodegradation		
Value	appr.	84	%
Duration		20	day(s)
Source	ECHA		
Evaluation	readily biodegradable		
3 **1-methoxy-2-propanol**	**107-98-2**		**203-539-1**
Type	aerobic biodegradation		
Value		96	%
Duration		28	day(s)
Method	OECD 301 E		
Source	ECHA		
Evaluation	readily biodegradable		
4 **ethyl-acetate**	**141-78-6**		**205-500-4**
Type	COD		
Value		60	%
Duration		10	day(s)
Source	ECHA		
Evaluation	readily biodegradable		

12.3 Bioaccumulative potential

Bioconcentration factor (BCF)		
No **Substance name**	**CAS no.**	**EC no.**
1 **ethyl-acetate**	**141-78-6**	**205-500-4**

BCF		30	
Source	ECHA		

Partition coefficient n-octanol/water (log value)			
No / **Substance name**		**CAS no.**	**EC no.**
1 / **butanone**		**78-93-3**	**201-159-0**
log Pow		0,3	
Reference temperature		40	°C
Method	OECD 117		
Source	ECHA		
2 / **ethanol**		**64-17-5**	**200-578-6**
log Pow		-0,35	
Reference temperature		24	°C
with reference to	pH 7,4		
Method	OECD 107		
Source	ECHA		
3 / **1-methoxy-2-propanol**		**107-98-2**	**203-539-1**
log Pow	<	1	
Reference temperature		20	°C
with reference to	pH: 6.8		
Method	OECD 117		
Source	ECHA		
4 / **ethyl-acetate**		**141-78-6**	**205-500-4**
log Pow		0,68	
Reference temperature		25	°C
with reference to	pH 7		
Method	EPA OPPTS 830.7560		
Source	ECHA		

12.4 Mobility in soil

No data available.

12.5 Results of PBT and vPvB assessment

Results of PBT and vPvB assessment	
Product Name	
74000-00199, ink, yellow	
PBT assessment	The product is not considered to be a PBT.
vPvB assessment	The product is not considered to be a vPvB.

12.6 Endocrine disrupting properties

Endocrine disrupting properties			
No	**Substance name**	**CAS no.**	**EC no.**
1	**sodium perchlorate**	**7601-89-0**	**231-511-9**
The substance has endocrine disrupting properties. (environment)			
Source ECHA			

12.7 Other adverse effects

No data available.

12.8 Other information

Other information
Do not discharge product unmonitored into the environment.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product

Disposal of the product should be carried out in accordance with all applicable regulations following consultation with the responsible local authority and the disposal company in an authorised and suitable disposal facility.
Allocation of a waste code number, according to the European Waste Catalogue, should be carried out in agreement with the regional waste disposal company.

Packaging
Used, completely emptied, packaging may be disposed of or passed to recovery systems in compliance with national and local provisions related to waste legislation. Like the unused product, the packaging that has not been emptied, may be disposed of or passed to recovery systems in compliance with national and local provisions related to waste legislation.

SECTION 14: Transport information

14.1 UN number or ID number

ADR/RID/ADN	UN1210
IMDG	UN1210
ICAO-TI / IATA	UN1210

14.2 UN proper shipping name

ADR/RID/ADN	PRINTING INK
IMDG	PRINTING INK
ICAO-TI / IATA	Printing ink

14.3 Transport hazard class(es)

ADR/RID/ADN - Class	3
Label	3
Classification code	F1
Tunnel restriction code	D/E
Hazard identification no.	33
Special Provision 640	640C
IMDG - Class	3
Label	3
ICAO-TI / IATA - Class	3
Label	3

14.4 Packing group

ADR/RID/ADN	II
IMDG	II
ICAO-TI / IATA	II

14.5 Environmental hazards

EmS	F-E, S-D

14.6 Special precautions for user
No data available.

14.7 Maritime transport in bulk according to IMO instruments
Not relevant

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU regulations

Regulation (EC) No 1907/2006 (REACH) Annex XIV (List of substances subject to authorisation)
According to the data available and/or specifications supplied by upstream suppliers, this product does not contain any substances considered as substances requiring authorisation as listed on Annex XIV of the REACH regulation (EC) 1907/2006.

REACH candidate list of substances of very high concern (SVHC) for authorisation
According to available data and the information provided by preliminary suppliers, the product does not contain substances that are considered substances meeting the criteria for inclusion in annex XIV (List of Substances Subject to Authorisation) as laid down in Article 57 and article 59 of REACH (EC) 1907/2006.

Regulation (EC) No 1907/2006 (REACH) Annex XVII: RESTRICTIONS ON THE MANUFACTURE, PLACING ON THE MARKET AND USE OF CERTAIN DANGEROUS SUBSTANCES, MIXTURES AND ARTICLES	
The product is considered being subject to REACH regulation (EC) 1907/2006 annex XVII.	No 3, 40
The product contains following substance(s) that are considered being subject to REACH regulation (EC) 1907/2006 annex XVII.	

No	Substance name	CAS no.	EC no.	No
1	butanone	78-93-3	201-159-0	75
2	ethyl-acetate	141-78-6	205-500-4	75
3	formaldehyde	50-00-0	200-001-8	75

Directive 2012/18/EU on the control of major-accident hazards involving dangerous substances	
This product is subject to Part I of Annex I, risk category:	P5b

Other regulations
Adhere to the national sanitary and occupational safety regulations when using this product.

National regulations

Water Hazard Class (Germany)

Class	1
Source	Classification according to AwSV (Regulation on facilities for handling substances that are hazardous to water).

15.2 Chemical safety assessment

A Chemical Safety Assessment has been carried out for one or more of the substances within this mixture.

SECTION 16: Other information

Sources of key data used to compile the data sheet:
Regulation (EC) No 1907/2006 (REACH), 1272/2008 (CLP) as amended in each case.
Directives 2000/39/EC, 2006/15/EC, 2009/161/EU, (EU) 2017/164.
National Threshold Limit Values of the corresponding countries as amended in each case.
Transport regulations according to ADR, RID, IMDG, IATA as amended in each case.
The data sources used to determine physical, toxic and ecotoxic data, are indicated directly in the corresponding section.

Full text of the H- and EUH- phrases drawn up in sections 2 and 3 (provided not already drawn up in these sections)

H226	Flammable liquid and vapour.
H271	May cause fire or explosion; strong oxidiser.
H302	Harmful if swallowed.
H373o	May cause damage to organs through prolonged or repeated exposure if swallowed.

Classification information
This product is assessed and classified using the methods and criteria below referred to in Article 9 of Regulation (EC) n° 1272/2008:
Physical hazards: determined through assessment data based on the methods or standards referred to in part 2 of Annex I to CLP
Health hazards and environmental hazards: determined through toxicological and ecotoxicological assessment data based on the methods or standards referred to in Part 3, 4 and 5 of Annex I to CLP.

Creation of the safety data sheet
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This information is based on our present knowledge and experience.
The safety data sheet describes products with a view to safety requirements.
It does not however, constitute a guarantee for any specific product properties and shall not establish a legally valid contractual relationship.

Alterations/supplements:
Alterations to the previous edition are marked in the left-hand margin.

This document is an English translation of the legally compliant safety data sheet of the region Germany. This document, including UFI and emergency telephone number, may only be used for placing on the market in the region Germany.

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